

ITM 31013

COMPUTER NETWORKS

Introduction



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COMPUTER NETWORK

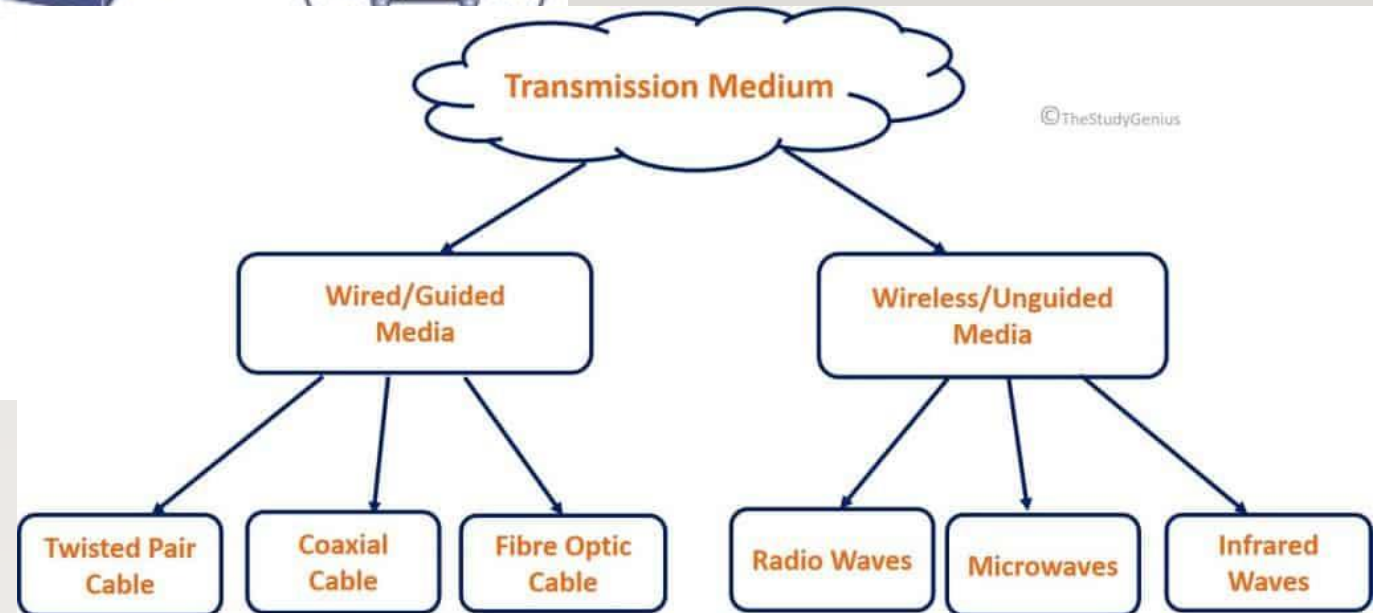
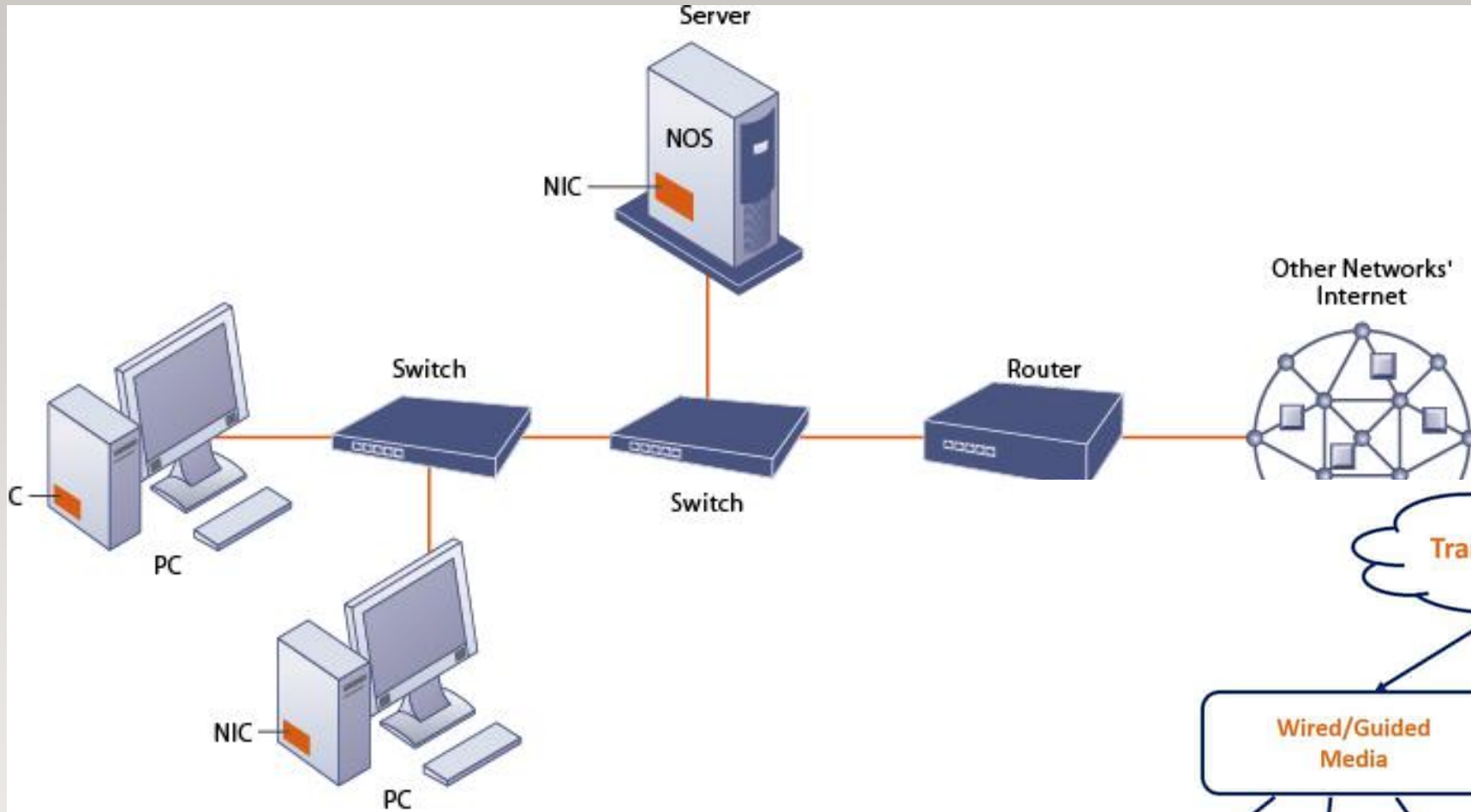


Computer network refers to the practice of connecting computers and other devices to share resources, information, and services. It involves the design, implementation, management, and maintenance of systems that link and enable communication between different computing devices. Networking can occur within a confined space, such as a single building or a campus (Local Area Network - LAN), or over larger geographical distances (Wide Area Network - WAN).

KEY COMPONENTS OF COMPUTER NETWORKING

1. **Devices:** Computers, servers, routers, switches, hubs, and other hardware components that enable communication.
2. **Media:** The physical or wireless medium through which data is transmitted, including Ethernet cables, fiber optics, and wireless signals.
3. **Protocols:** Sets of rules and conventions that govern communication between devices, ensuring compatibility and reliability.
4. **Topologies:** The arrangement or structure of the network, determining how devices are interconnected (e.g., star, bus, ring, mesh).

KEY COMPONENTS OF COMPUTER NETWORKING



PURPOSE OF COMPUTER NETWORKING

- **Resource Sharing:** Networks allow sharing of resources such as files, printers, and internet connections, promoting efficiency and collaboration.
- **Communication:** Facilitates communication between users, devices, and systems, enabling the exchange of information and data.
- **Centralized Management:** Centralized control and management of resources, security, and access rights.
- **Remote Access:** Enables remote access to resources and services, providing flexibility and accessibility.

COMMUNICATION

Communication in a general sense refers to the process of **exchanging information**, ideas, thoughts, or feelings between individuals or groups. It is a fundamental aspect of human interaction and a crucial element in various forms of life. Communication can take place through verbal or non-verbal means, using language, gestures, symbols, or other mediums to convey messages.

KEY ELEMENTS OF COMMUNICATION

1. Sender:

The person or entity initiating the communication by encoding a message. This can be an individual, group, organization, or even a machine.

2. Message:

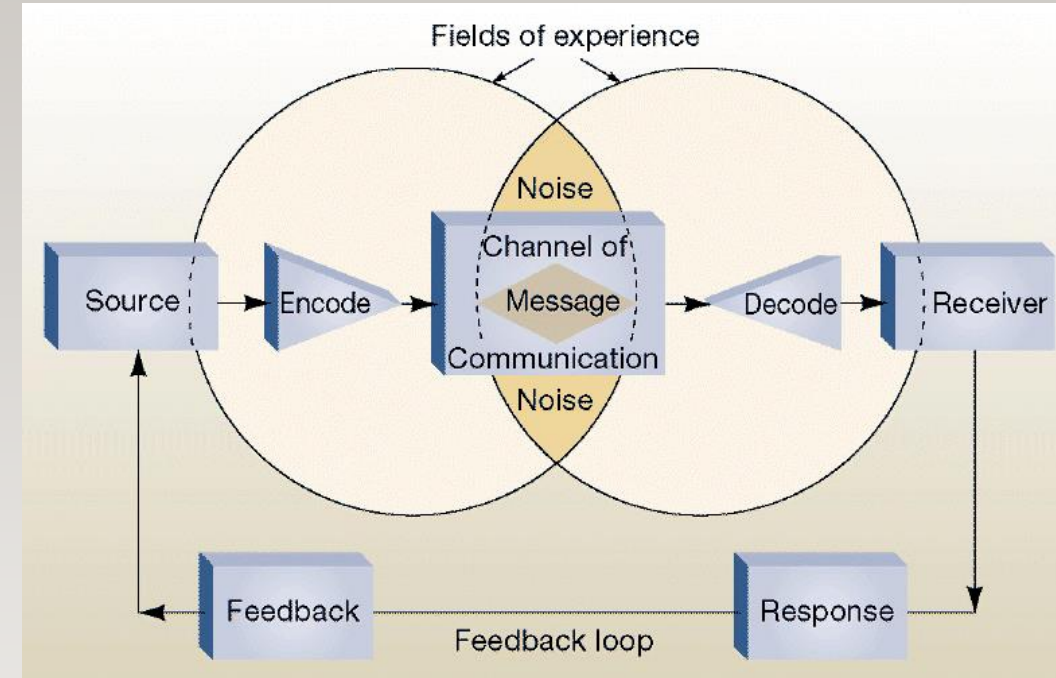
The information, idea, or emotion being conveyed by the sender. Messages can take various forms, including spoken or written words, images, signals, or any other means of expression.

3. Channel:

The medium or method through which the message is transmitted. This can include face-to-face conversations, written documents, emails, phone calls, video conferences, and more.

4. Receiver:

The individual or group for whom the message is intended. Receivers decode the message to understand its meaning. In the case of feedback, the receiver can become the sender, initiating a new cycle of communication.



KEY ELEMENTS OF COMMUNICATION

5. Feedback:

The response or reaction of the receiver to the sender's message. Feedback is crucial for effective communication as it allows the sender to gauge the understanding and interpretation of the message.

6. Context:

The circumstances or environment in which the communication takes place. Context includes the physical setting, cultural background, social relationships, and any other factors that may influence the interpretation of the message.

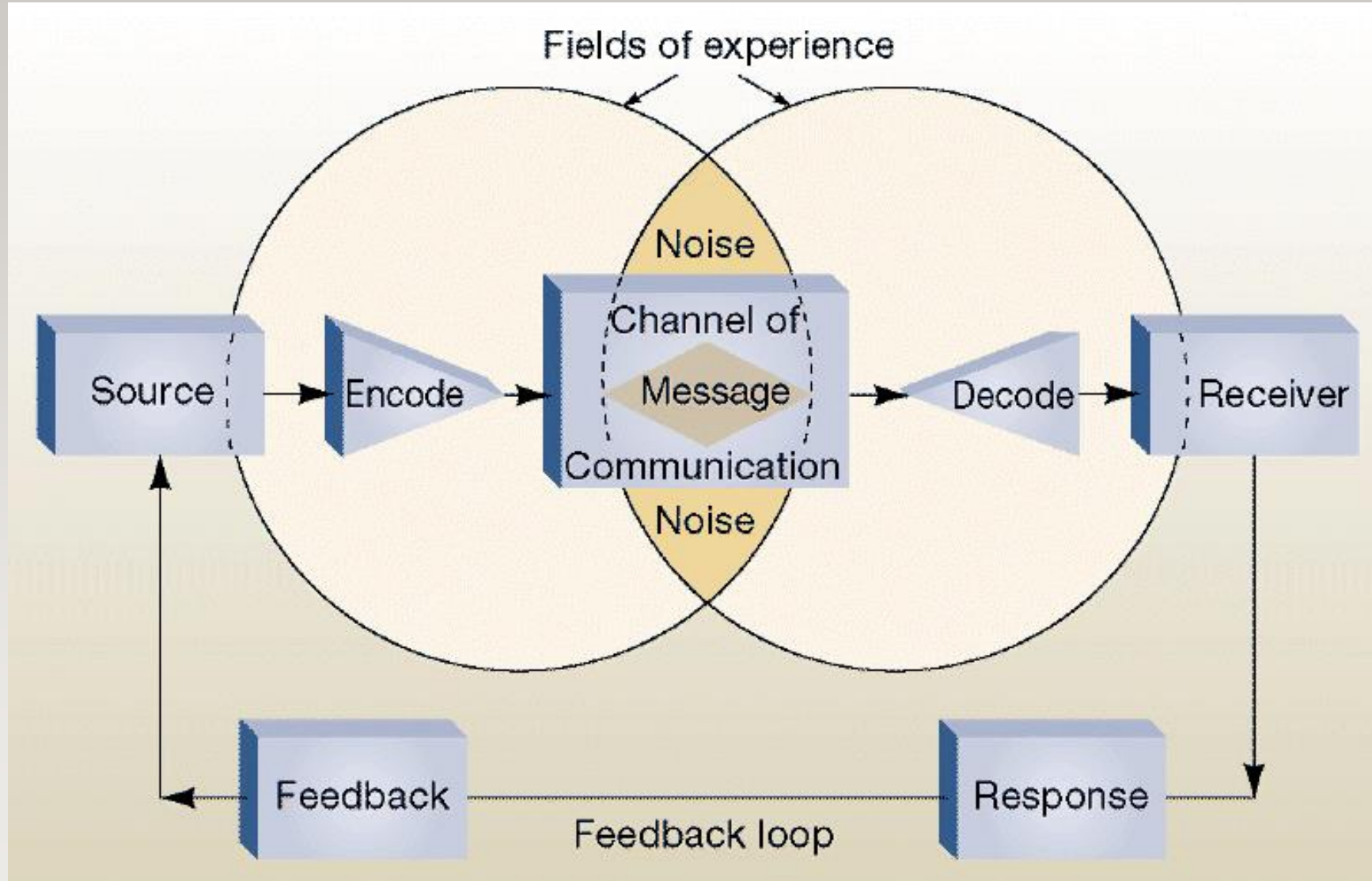
7. Noise:

Anything that interferes with the successful transmission or reception of a message. Noise can be physical (background sounds), semantic (language barriers), or psychological (preconceived notions).

8. Purpose:

Communication often serves a specific purpose, such as informing, persuading, entertaining, or expressing emotions. Understanding the purpose helps shape the message and the choice of communication channels.

ABSTRACT MODEL OF COMMUNICATIONS



ABSTRACT MODEL OF COMMUNICATIONS

1. Sender:

- Definition: The individual or entity initiating the communication by encoding a message and transmitting it to the receiver.
- Example: An employee drafting an email to inform colleagues about an upcoming project meeting.
- Real-World Scenario: Alice, a project manager, sends an email to her team members to schedule a meeting to discuss the project timeline and deliverables.

2. Message:

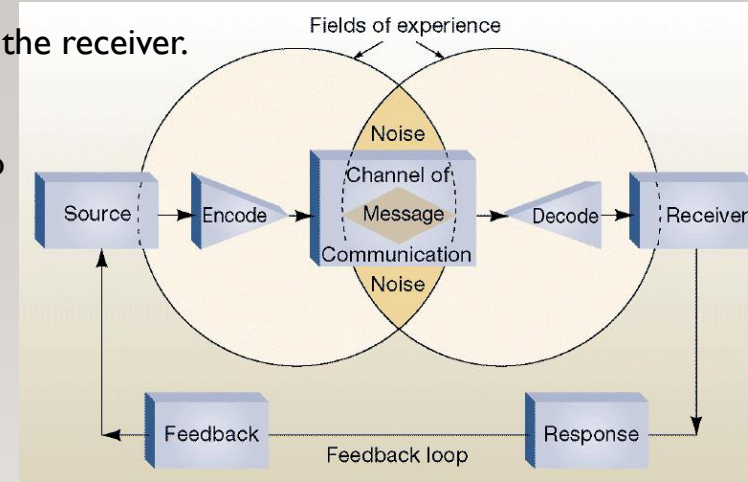
- Definition: The information, idea, or content being conveyed by the sender to the receiver.
- Example: The text, images, or multimedia content contained within the email sent by the employee.
- Real-World Scenario: In Alice's email, the message includes details about the meeting agenda, date, time, and location, along with any relevant attachments or links.

3. Channel:

- Definition: The medium or method used to transmit the message from the sender to the receiver.
- Example: Email, phone call, face-to-face conversation, video conference, or written letter.
- Real-World Scenario: Alice chooses email as the channel to communicate with her team due to its convenience and ability to reach all members simultaneously.

4. Encoding:

- Definition: The process of converting the message into a format suitable for transmission through the chosen communication channel.
- Example: Converting the text of the email into digital data packets for transmission over the internet.
- Real-World Scenario: Before sending the email, Alice composes the message using her email client, which converts the text into electronic data for transmission.



ABSTRACT MODEL OF COMMUNICATIONS

5. Noise:

- Definition: Any interference or disturbance that affects the transmission or reception of the message, potentially leading to misinterpretation or loss of information.
- Example: Background noise during a phone call, distractions in the environment, or technical issues with the communication channel.
- Real-World Scenario: While Alice is on the phone discussing project details with a team member, construction noise outside her office creates interference, making it difficult to hear clearly.

6. Decoding:

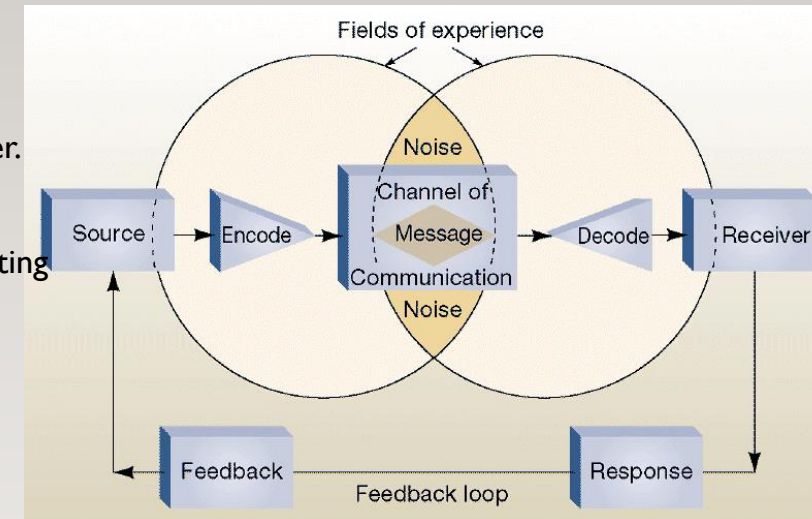
- Definition: The process by which the receiver interprets and understands the message received from the sender.
- Example: Reading and comprehending the content of the email received from the sender.
- Real-World Scenario: After receiving Alice's email, each team member reads the message, understands the meeting details, and prepares accordingly for the upcoming discussion.

7. Receiver:

- Definition: The individual or entity intended to receive and interpret the message transmitted by the sender.
- Example: The team members who receive and read the email sent by the sender.
- Real-World Scenario: Each member of Alice's team acts as a receiver when they receive and process the email communication regarding the project meeting.

8. Feedback:

- Definition: The response or reaction provided by the receiver to the sender, indicating their understanding or acknowledgment of the message.
- Example: Reply to an email, acknowledgment nod during a conversation, or verbal confirmation over the phone.
- Real-World Scenario: After reading Alice's email, team members reply with confirmation of their attendance, providing feedback to the sender regarding their understanding and acknowledgment of the meeting details.



DIRECTION OF DATAFLOW

1. Simplex Communication:

- Simplex communication is a unidirectional mode where data flows in only one direction, from the sender to the receiver, with no feedback or response. → Example: Television / Radio broadcast, one-way paging systems.
- Scenario: Watching a TV program without any interaction or feedback from the viewers.

2. Half-Duplex Communication:

- Definition: Half-duplex communication allows data transmission in both directions, but not simultaneously. Data can be sent and received, but not at the same time. → Example: Walkie-talkies, push-to-talk (PTT) communication systems.
- Scenario: Police officers using walkie-talkies can transmit messages to headquarters and receive responses, but they cannot speak and listen simultaneously.

3. Duplex Communication:

- Definition: Duplex communication enables simultaneous two-way data transmission between the sender and the receiver, allowing communication in both directions at the same time. → Example: Telephone conversations, video conferencing.
- Scenario: During a phone call, both parties can speak and listen simultaneously, allowing for a natural conversation flow without the need to take turns transmitting messages.

BENEFITS OF USING COMPUTER NETWORKS

a) Communication

- **Example:** Email, video conferencing, and instant messaging enable people worldwide to communicate efficiently.

b) Resource Sharing

Share hardware devices like printers and storage.

- **Example:** Multiple computers in an office accessing a shared printer.

c) Data Sharing

Allows files, documents, and applications to be accessed and modified by multiple users.

- **Example:** Cloud storage services like Google Drive.

d) Internet Access

Shared network infrastructure for internet connectivity.

- **Example:** Homes or offices using a single router for internet.
- **Example:** Employees working from home via a VPN.

BENEFITS OF USING COMPUTER NETWORKS

e) Cost Efficiency

Reduces costs by sharing resources instead of buying separate devices for each user.

- **Example:** A small business shares a high-speed internet connection among employees.

f) Enhanced Security

Centralized control allows easier implementation of security policies.

- **Example:** A company restricts access to confidential files using a firewall.

g) Scalability

Networks can be expanded easily by adding devices.

- **Example:** Adding new computers to an office network.

h) Remote Access

Allows users to access resources from anywhere.

Example: Employees working from home via a VPN.

BASIC COMPONENTS OF COMPUTER NETWORKS

1. Physical Components
2. Logical Components
3. Software Components
4. People as Components
5. Networking Services
6. Additional Tools

BASIC COMPONENTS OF COMPUTER NETWORKS

I. Physical Components: These are tangible parts of a computer network.

a) End Devices:

Devices that directly interact with the user or generate data.

- **Examples:** Computers, laptops, smartphones, tablets, IoT devices.

b) Networking Devices

Devices that facilitate communication between end devices and manage traffic.

- **Switches:** Connect devices in a LAN and use MAC addresses for forwarding.
- **Routers:** Connect different networks and use IP addresses to route traffic.
- **Hubs:** Simple devices that broadcast data to all connected devices.
- **Bridges:** Connect two or more LANs.
- **Access Points (APs):** Provide wireless connectivity within a network.
- **Gateways:** Connect two different types of networks, such as a LAN and the internet.
- **Modems:** Convert digital signals to analog (and vice versa) for internet access.

BASIC COMPONENTS OF COMPUTER NETWORKS

c) **Transmission Media:** The medium through which data is transmitted.

- **Wired Media:**
 - Twisted Pair Cable (e.g., Ethernet cables).
 - Coaxial Cable.
 - Fiber Optic Cable (high-speed and long-distance).
- **Wireless Media:**
 - Radio Waves (Wi-Fi).
 - Infrared.
 - Satellite Communications (e.g., Starlink).
 - Cellular Networks (4G, 5G).

d) **Servers:** Centralized systems that provide resources or services to other devices.

- **File Servers:** Store and manage files.
- **Database Servers:** Manage database queries.
- **Web Servers:** Host websites.
- **Email Servers:** Handle email communications.

BASIC COMPONENTS OF COMPUTER NETWORKS

2. Logical Components : These are intangible yet essential for enabling communication and data exchange.

a) Protocols: Rules that govern communication between devices in a network. They ensure standardized and reliable data transfer.

- **Examples:**

- **HTTP/HTTPS:** For web browsing.
- **FTP:** For file transfer.
- **SMTP/IMAP/POP3:** For email.
- **TCP/IP:** Core protocol for communication over the internet.
- **DNS:** Resolves domain names to IP addresses.
- **DHCP:** Assigns IP addresses dynamically.

b) Addressing Mechanisms: Assign unique identifiers to devices for communication.

- **MAC Address:** Unique hardware address for devices in the same network.
- **IP Address:** Logical address for devices (IPv4 or IPv6).

c) Network Topologies: The physical or logical layout of a network.

- **Examples:** Star, Bus, Ring, Mesh, Hybrid.

BASIC COMPONENTS OF COMPUTER NETWORKS

3. Software Components

Software facilitates network management, configuration, and functionality.

a) Operating Systems

- Manage the underlying hardware and enable network functionality.
 - **Examples:** Windows Server, Linux (e.g., Ubuntu), macOS.

b) Network Management Tools

- Monitor and manage network resources.
 - **Examples:** Wireshark, SolarWinds, Nagios.

c) Firewalls and Security Software

- Protect networks against unauthorized access and threats.
 - **Examples:** pfSense, ZoneAlarm, Sophos.

BASIC COMPONENTS OF COMPUTER NETWORKS

4. People as Components

- Humans are often overlooked as components, but they are integral to networks.
- **a) Network Administrators**
 - Manage, configure, and troubleshoot the network.
- **b) Users**
 - Access network resources, generate traffic, and provide feedback.
- **c) Security Teams**
 - Implement security measures and ensure network integrity.

BASIC COMPONENTS OF COMPUTER NETWORKS

5. Networking Services

These provide the functionality users rely on.

a) Directory Services

Manage user access and permissions.

- **Examples:**Active Directory.

b) Virtualization

Enables the creation of virtual networks or machines.

- **Examples:**VMware,VirtualBox.

c) Cloud Services

Provide scalable resources over the internet.

- **Examples:**AWS, Google Cloud.

BASIC COMPONENTS OF COMPUTER NETWORKS

6. Additional Tools

Accessories that enhance or extend network functionality.

a) Repeaters

Amplify signals for long-distance data transmission.

b) Load Balancers

Distribute traffic across multiple servers.

c) Intrusion Detection Systems (IDS)

Monitor and detect potential security threats.